

ANDREW NGUYEN

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Education

University of Michigan

Aug. 2023 – May 2027

B.S. in Computer Science & B.S. in Robotics, Entr. Minor, Eng. Honors Program, GPA: 3.8/4.0

Ann Arbor, MI

Coursework: ML, RL, Bayesian ML, Algorithms for Robotics, Robot Controls, ROS, SLAM & Navigation

Experience

Biomimetic Robotics Lab

May 2026 – Present

Incoming Visiting Student

Cambridge, MA

ARCAD Lab (Project with Schaeffler)

Sep. 2025 – Present

Robotics Simulation Engineer — Python, MuJoCo, NumPy, PyTorch, RL, ROS, Git

Ann Arbor, MI

- Integrating actuator dynamics and RL locomotion policy into MuJoCo humanoid sim to bridge Sim to Real gap; validating fidelity using hardware dyno data
- Developing hybrid actuator models by synthesizing analytical physics-based equations with Neural Networks to capture complex non-linear behaviors

FREE Lab

Aug. 2023 – Present

Undergraduate Research Assistant — Python, PyBullet, Git, Onshape

Ann Arbor, MI

- Modeling interconnected fixed-wing UAVs for increased efficiency and maneuverability using PyBullet
- Designed and prototyped passive wall-climbing robot using bi-stable suction; successfully scaled vertical plaster & glass surfaces carrying 8kg payload (first-author paper accepted to ICRA 2026)
- *Outstanding Presentation Award*, Gulf Coast Undergraduate Research Symposium, Rice University, November 2024

RAI Institute (Formerly Boston Dynamics AI Institute)

May – Aug. 2025

Hardware Intern — Python, Git, Onshape, Linux

Cambridge, MA

- Executed several critical-path tasks for a project to advance state-of-the-art robotics performance
- Wrote automated motor dynamometer test suite in Python to validate actuator builds
- Analyzed coulomb friction, viscous, and torque-velocity plots of custom actuators, adjusting assembly as necessary

RAI Institute (Formerly Boston Dynamics AI Institute)

May – Aug. 2024

Hardware Intern — Onshape

Cambridge, MA

- Designed 1-DOF camera mount, in Onshape, for bimanual robot manipulator
- Assembled 40 quasi-direct-drive actuators using personally designed test fixtures (Onshape, CNC, metal lathe)

Publications

- [1] A. Nguyen, M. Li, and D. Bruder, “A high-payload wall-climbing robot using passive bistable suction cups,” in *2026 IEEE International Conference on Robotics and Automation (ICRA)*, Accepted for publication, Vienna, Austria, 2026.

Projects

The Brainstormers FTC Team 8644

2012 – 2023

Team Captain, Head Robot Designer, Software Lead — Java, Git, Fusion 360, OpenCV

Lexington, MA

- Led engineering and software for competitive robotics team; wrote 4,000+ lines of Java control and OpenCV vision
- 2nd Place World Championship (2022), 6x State Champion; robot ranked #1 globally (2023), World Champion (2018)
- Mentored 100+ teams and presented work to MA Lt. Governor and corporate partners

Technical Skills

Languages: C++, Python, ROS, C, Java, HTML/CSS, JavaScript, Julia

Libraries: NumPy, Pandas, Matplotlib, PyTorch, MuJoCo, PyBullet

Fabrication: 3D Printing, Machine Shop Tools, CNC Routing, Metal Lathe, Silicon Molding

Applications: Onshape, SolidWorks, Autodesk Fusion 360, 3D Printer Slicers

Tools: Git, Linux, CI/CD, Unit Testing, Jupyter, OpenCV